

Diffuse the Object, Keep Its Label: Curating Detector Training Data from a Few Unlabeled Photographs via VLM-Built 3D Vegetation Scenes

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Abstract. Labeled images of small objects hidden in vegetation are scarce, and detectors trained on them generalize poorly across sites. Rather than reusing labels collected at another site, we synthesize labeled training images from a handful of unlabeled photographs of the deployment site itself. A vision-language model generates a coarse 3D vegetation scene from one photograph; placing 3D object meshes in the scene yields bounding boxes, segmentation masks, and per-instance occlusion directly from the scene geometry, without manual annotation. A lightweight adapter fine-tuned on the photographs conditions a diffusion pass that re-textures the renders, and a graded mask-lock sets how much diffusion may touch the object itself. In our runs this grade was the most influential curation choice: lightly diffusing the object improves minority-class recall over fully protecting its pixels, while unrestricted diffusion dissolves it. Trained on these images, a standard detector matched or exceeded its counterpart trained on a larger labeled dataset of real images from a different site, consistently across seeds on a humanitarian-demining benchmark; the comparison is thus unsupervised site adaptation from a handful of photographs against conventional cross-site label reuse. In our ablations the gains were largely insensitive to the photograph and crop budgets, and in-domain accuracy did not predict cross-site performance.

Keywords: Synthetic training data · Diffusion models · Humanitarian demining

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1 Introduction

Detecting small or camouflaged objects in natural vegetation is a recurring problem across safety-critical and resource-constrained domains, from humanitarian landmine clearance [9, 23, 40] to invasive species monitoring [5], precision agriculture [35], and wilderness search and rescue [16]. These tasks share a bottleneck: annotated data are scarce, expensive to collect, and often physically hazardous to acquire.

Synthetic images are a natural response to this scarcity, but existing approaches have well-documented limits. Procedural 3D rendering in open-source engines such as Blender [4], used for detector training via domain randomization [30, 37], offers full control over object pose, occlusion, and label geometry, yet without extensive artist input its output looks synthetic: flat foliage textures, uniform shading, and no sensor noise. Neural reconstructions based on 3D Gaussian Splatting [39, 42] produce visually convincing backgrounds but require dense captures of the target environment, which are often unavailable in the domains listed above. Text-to-image diffusion models [32] yield photorealistic outputs from prompts but do not natively expose the pose and label information needed for detector training, nor can they exploit an available 3D model of the object.

A promising middle ground is to render a coarse 3D scene and then refine its appearance with a diffusion model. Recent instances of this pattern pair Blender rendering with a depth-conditioned diffusion model for mushroom segmentation [15], refine Skinned Multi-Person Linear (SMPL) human renders with an edge-conditioned Stable Diffusion model for privacy-compliant human synthesis [27], and insert 3D vehicle assets into Gaussian-splatting scenes via a one-step diffusion model for driving simulation [21]. IntrinsicControlNet [22] conditions generation on albedo, normal, and lighting buffers extracted from a renderer, and Coarse-to-Real [8] extends the idea to game-engine crowd videos. In parallel, LLM- and VLM-driven agents synthesize executable Blender code as a scene scaffold that a diffusion model can then refine: SceneCraft [12] translates a text description into Blender code through scene-graph planning, and SEIG [10] goes further by reconstructing a Blender scene from a single reference photograph via staged inverse graphics.

We build on both lines and address a deployment setting common in humanitarian mine action: the operator knows, from a non-technical survey [14] and historical clearance records [33], both the terrain type of the suspected hazardous area and the family of ordnance likely to be encountered. This prior knowledge, standard practice in mine action doctrine, makes a site-specific synthetic dataset feasible without large-scale per-site data collection: the terrain distribution can be captured by a few unlabeled reference photographs, and the object catalog reduces to a handful of known mesh models. The conventional alternative is to reuse labeled data collected at previously surveyed sites, and it is exactly this cross-site reuse that breaks down: every new suspected hazardous area is, in effect, a new visual domain for which labels do not exist. Generating the training set from a few photographs of the site itself sidesteps the reuse problem and can be repeated for each new site at negligible data-collection cost.

Our pipeline proceeds as follows. A VLM reads one reference photograph and emits a coarse Blender scene of the terrain; the scene is populated with 3D meshes of the expected ordnance (in our study, four PFM-1 butterfly landmine variants and one PMA-2 starfish anti-personnel landmine), so every render carries bounding boxes, masks, and per-instance occlusion by construction. A Low-Rank Adaptation (LoRA) [11], fine-tuned on unlabeled crops of the target terrain, then guides SDEdit [25] to replace surface appearance while preserving the 3D layout, and a sensor-noise pass matches the target camera response.

Existing render-then-refine pipelines assume that the diffusion model can plausibly re-texture the object; this holds for mushrooms, vehicles, and human bodies, but becomes fragile for objects that are rare or absent from generative training corpora, such as ordnance: unconstrained diffusion can distort, remove, or hallucinate around the object and invalidate the associated label. We enforce preservation directly through *mask-locked compositing*: the diffusion output is used only for the background, while the object pixels rendered by Blender are pasted back through the render’s visible-surface (post-occlusion) mask, so occluding vegetation stays in front of the object after the paste. The label therefore remains valid by construction.

A further property separates this setting from prior render-and-refine applications: surface-laid objects often lie partially *embedded in* vegetation rather than solely on top of it. Occlusion by vegetation is not a marginal effect but a primary driver of detection failure in minefield environments [2], and a generative model alone cannot reproduce it with the supervision detection requires: diffusion exposes neither the visible/amodal mask split nor the coverage ratio, and thin occluders such as grass blades fall below the effective resolution of latent-space conditioning. In our pipeline, occlusion is instead inherited from the 3D scene: the vegetation primitives generated by the VLM geometrically cover the object, the visible-surface mask is exact, and occlusion is controlled per instance.

We instantiate the pipeline on surface-laid landmine detection in vegetated terrain, a data-scarce and safety-critical detection task in computer vision. Existing benchmarks [23, 40] together provide only a few tens of thousands of annotated images across a handful of terrain types, and expanding them requires inert ordnance and controlled field sites. The pipeline is not tied to this task in principle: other detection settings in which the object catalog is small, geometrically well defined, and hard to acquire in real data could substitute their own 3D mesh library and terrain LoRA, though we do not evaluate that generalization here.

Contributions.

1. A render-and-refine pipeline for curating detector training data in vegetated terrain: a VLM generates a coarse Blender scene from one reference photograph, and a LoRA trained on unlabeled crops of the same site adapts the diffusion prior. A single graded mask-lock parameter α acts as a curation dial: it sets how much diffusion may touch the object, while mask-locked compositing preserves the placed 3D meshes and their labels.

2. A preliminary cross-site generalization study on surface-laid landmine detection against the same detector trained on a far larger labeled dataset of real images: in our runs, varying the mask-lock grade alone shifted generalization more than any data-budget choice (Sec. 4).

2 Related Work

Synthetic data for object detection. Generating training images from 3D engines is a well-established strategy. Domain randomization varies lighting, texture, and geometry during rendering to broaden the effective support of the training distribution [36, 37]. Structured variants impose scene-level priors such as object placement rules and context-aware composition [30]. A complementary line replaces the renderer with a 2D generative model and relies on compositing for the background. In agriculture, Generate-Paste-Blend-Detect [7] generates whitefly instances with a denoising diffusion probabilistic model (DDPM) and composites them onto real backgrounds using Poisson or Gaussian blending to train YOLOv8. Modak *et al.* [26] apply Stable Diffusion inpainting to augment weed images under model quantization constraints, showing that generative augmentation remains effective on edge hardware. Patch-level synthesis [18] composites species-level crop and weed instances onto real backgrounds for fine-grained segmentation. These 2D approaches produce plausible images but lack geometric grounding: object scale, occlusion, and viewpoint are controlled only implicitly, and annotations must be derived from the generation mask rather than from known 3D geometry.

3D rendering with diffusion refinement. A growing line of work addresses the appearance gap between renders and real images by rendering a coarse 3D scene and then transforming it into a photorealistic image with a diffusion model. The closest system to ours is that of Károly and Galambos [15], who regenerate entire Blender-rendered mushroom scenes with depth-conditioned Stable Diffusion XL (SDXL), reaching $F1 = 0.859$ on the M18K benchmark with only synthetic training data. Full regeneration is viable in their setting because the target class is well represented in generative corpora (their object LoRA is trained largely on web images), yet they report that the depth conditioning occasionally admits unlabeled instances that bias the trained model toward precision over recall. Notably, on their own real-world data the diffusion-refined training set does not outperform the plain Blender renders, leaving open whether generative refinement helps precisely where it should matter most: under domain shift. RefSD [27] pseudonymizes humans by rendering an SMPL mesh over each subject and passing it through SDXL with a Canny ControlNet, so the 3D scaffold preserves pose while the diffusion redraws surface appearance under text-prompt attribute control. R3D2 [21] takes a different route: it inserts 3D vehicle assets into Gaussian-splatting reconstructions of driving scenes and uses a one-step diffusion model to add shadows and lighting integration rather than to re-texture the scene. Coarse-to-Real [8] extends the render-then-refine idea to

video, learning a neural renderer that transforms game-engine crowd simulations into photorealistic sequences. IntrinsicControlNet [22] conditions generation on intrinsic buffers (albedo, normals, lighting) extracted from a rendering engine, bridging synthetic and real distributions in a single model.

A common assumption across these systems is that the diffusion model can plausibly re-texture the target category: mushrooms, vehicles, and human bodies are all well represented in generative training corpora. We consider the harder case in which the target object has no meaningful generative prior, and preserve it explicitly through mask-locked compositing against the visible-surface mask of the render. The two designs trade off differently: unconstrained generation can corrupt labels by adding or altering instances [15], whereas locked compositing keeps labels exact and concentrates any residual appearance gap on the object itself, whose realism is then governed by the fidelity of its rendered texture, a trade-off we examine in Section 4.

Gaussian splatting and VLM-driven scene construction. An alternative to procedural rendering is to reconstruct a scene from real captures using 3D Gaussian Splatting. Cut-and-Splat [39] trains a Gaussian-splatting model of a target object from a video, automatically extracts the object, and re-renders it onto random backgrounds with a depth-guided pose, outperforming cut-and-paste and diffusion baselines on their instance-segmentation benchmark. Zanjani *et al.* [42] use splatting to insert 3D objects into reconstructed driving scenes for 3D detection. These methods are visually strong but require dense captures of the scene, which are rarely available in obscure detection domains. On the scene-specification side, SceneCraft [12] and SEIG [10] show that LLM and VLM agents can synthesize executable Blender code from a text description or a single image respectively, via scene-graph planning with iterative library learning or staged inverse graphics. We use a similar VLM-to-Blender step as the entry point to our pipeline, and defer photorealism to the subsequent LoRA-conditioned diffusion pass rather than to the renderer.

Landmine detection and data scarcity. Surface-laid anti-personnel landmines pose acute detection challenges due to their small size, camouflaging coloration, and high variability across terrain and vegetation conditions [23], motivating robotic multi-sensor platforms for their detection and disposal [9]. The MineInsight dataset [23] covers three off-road vegetated tracks containing 15 inert landmines and 20 distractor items, with synchronized monochrome, RGB, VIS-SWIR, LWIR (thermal), and LiDAR streams from a UGV-mounted robotic arm, while SULAND [40] offers ground-level RGB images of PFM-1 (butterfly) and PMA-2 (starfish) landmines across Italian and US sites. Together, these benchmarks provide on the order of a few tens of thousands of annotated frames and cover only a handful of terrain types, far below the diversity needed for reliable generalization. Vegetation height and density further degrade detection accuracy, as quantified by Baur *et al.* [2] for unmanned aerial vehicle (UAV) sensing in minefield environments. Collecting additional real data requires controlled environments with inert ordnance and extensive safety protocols, making large-scale annotation pro-

hibitively expensive; even thermal detection of the PFM-1 in vegetation operates under severely limited training data [24]. Synthetic generation for this task has so far been limited to 2D cut-and-paste composition of landmine templates onto real backgrounds [1, 31], which can neither reproduce the photometric coupling between a landmine and its surroundings (cast shadows, ambient occlusion, inter-reflection) nor embed the target geometrically *within* the vegetation, precisely the property identified above as decisive for detection [2].

Copy-paste and compositing augmentation. Copy-paste augmentation [6] randomly composites object instances onto new backgrounds and has become a standard augmentation in instance segmentation. Extensions apply Poisson blending [28] or diffusion-based harmonization [34, 41] to reduce boundary artifacts. Cut-and-paste strategies have been adopted for illegal landfill detection [17] and other environmental monitoring tasks where object appearance is stable but background diversity is limited. Our mask-locked compositing shares the motivation of preserving the original object appearance, but operates in 3D: the occluding vegetation is geometrically placed, the mask is exact, and the label derives from the scene graph rather than from a 2D segmentation.

3 Method

Figure 1 summarizes the pipeline; we describe each stage below.

Stage 1: VLM scene primitives. We use a commercial vision–language model as the scene generator. Prompted with one reference photograph, the VLM directly emits Python code that instantiates the terrain in Blender as simple primitives: ground palette, vegetation cards, and clutter (rocks, sticks, patches of bare soil). No pre-owned asset library is involved; at this stage the scene consists of minimal primitives rather than a full reconstruction, and the camera is subsequently sampled near the initial camera pose estimated by the VLM. 3D meshes of the target ordnance are then placed by a scripted spawner: several variants of the PFM-1 butterfly landmine and the PMA-2 anti-personnel landmine, sampled under a balanced class draw with randomized yaw, position, and partial burial. The spawner is fully parametric: target positions, class distribution, spatial clustering, and burial can all be altered at will, so the same scene supports arbitrary target layouts. Scene lighting is likewise not hand-set: per-scene sun strength and exposure are derived from the mean luminance of the reference photographs; richer parameterizations (time of day, sun position) are left to future work. We deliberately do not ablate the VLM choice. Commercial VLMs evolve on a shorter cadence than a research pipeline, so a fixed-version claim would age within months; we treat the VLM as a swappable scene-generation backend and report the prompt template in Appendix A.

Stage 2: adaptation pre-phase, unlabeled crops and terrain LoRA. This stage runs once per site, before any image generation; the later stages only apply its

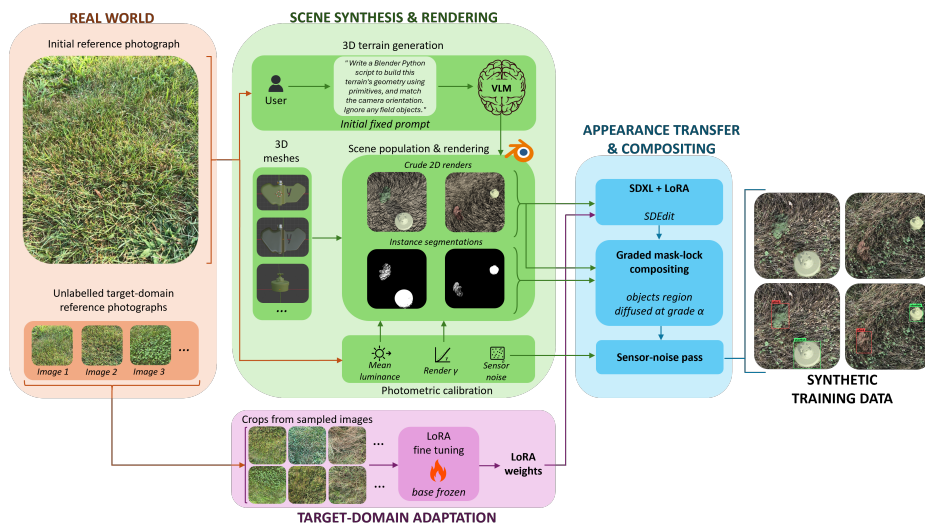


Fig. 1: Overview of the proposed pipeline. The real-data budget is a handful of unlabeled photographs of the target site. *One* of them serves as the scene reference: a vision–language model emits a Blender Python script that approximates the terrain geometry from primitives (SCENE SYNTHESIS & RENDERING). 3D meshes of the target ordnance are placed in the resulting scene, so bounding boxes, instance masks, and per-instance visibility are obtained directly from the scene graph without manual annotation. The *full* photograph set feeds a one-time adaptation pre-phase, prepared once per site and reused for every generated image: sun strength, render gamma, and sensor noise are calibrated on the photographs, and a LoRA fine-tuned on crops sampled from them adapts the frozen diffusion prior to the target domain (TARGET-DOMAIN ADAPTATION). Each crude render is then re-textured by an SDEdit pass under the LoRA-adapted prior, while a second copy bypasses diffusion and rejoins at graded mask-lock compositing, where the object region is blended at grade $\alpha \in [0, 1]$; a final capture-realism pass matches the target camera response. Labels therefore remain valid by construction. Boxes in the output denote the two target classes.

output. The pipeline’s entire real-data budget is a handful of unlabeled photographs, sampled from the existing distribution of the SULAND dataset in the spirit of a generic robot mission over a field; the scene reference of Stage 1 is one of them. This rests on the working hypothesis that the field’s appearance stays approximately stationary across the site, with near-constant vegetation; visually heterogeneous sites would require reference photographs per appearance mode. We take random square crops of each photograph, without bounding boxes, masks, or class labels of any kind. On these crops we fine-tune a LoRA adapter [11] on a frozen Stable Diffusion XL backbone [29]. The caption is a fixed short description of the target terrain and is reused verbatim at inference. The adapter is intentionally small so that it captures the terrain’s surface statistics

rather than individual images; its configuration is fixed across all experiments and reported in Appendix A.

Stage 3: mask-locked appearance transfer. We render one balanced pool of 1,000 scenes and reuse it for every ablation: the targets spawned in the 3D scene, their poses, and their occlusions remain identical across all cells, so scene variance is not a confound. Passing through a 3D engine is what provides generic coverage of the object in 3D and, importantly, full control over occlusion of the target by vegetation at the initial stage, a property we consider central to applications such as landmine detection and target identification in vegetation. Each render is passed through SDEdit [25], prompted with the LoRA caption. SDEdit re-noises the render into the diffusion latent space and denoises it under the LoRA-conditioned prior, transferring surface appearance while the 3D layout constrains structure. For labels to remain valid, we then paste the rendered object pixels back through the visible-surface mask exported by Blender. We control how much diffusion may touch the object itself through a graded mask-lock parameter $\alpha \in [0, 1]$: at $\alpha=0$ the object is pixel-protected, at $\alpha=1$ the diffusion output is kept everywhere. Intermediate values feather the mask so that a thin, controllable share of the object’s appearance is inherited from the diffusion pass while its geometry and label remain fixed. In our experiments this grade emerges as an influential curation choice (Section 4). Figure 2 shows representative outputs across α .

Stage 4: sensor-noise pass. The composited image is finally passed through a light camera model: we estimate the sensor-noise level of the reference photographs [13] and inject noise so that the generated images approximate the measured statistic; a fixed compression pass completes the model.

Stage 5: detector training. We train an off-the-shelf YOLOv11-L detector [38], initialized from its COCO-pretrained checkpoint [19], on the generated dataset. The training recipe follows the SULAND benchmark [40] one-to-one, so the synthetic-versus-real comparison in Section 4 isolates the training data as the only variable. One consequence of rendering from a single scene deserves note: because all generated images share the same background and asset combinations, any validation split of them stays in-distribution, and in-domain validation cannot indicate when a model is ready to generalize (Sec. 4).

The complete generation and training parameters, including the SULAND augmentation policy, are tabulated in Appendix A together with the VLM prompt template.

4 Experiments

The experiments address three points: (i) whether the pipeline generalizes to real images, and how it compares against a baseline trained on real labeled data under the same training procedure (Sec. 4.2); (ii) which curation choice governs

generalization: how much data is used, or how the object itself is processed (Sec. 4.3); (iii) whether anything cheaper than downstream evaluation predicts generalization (Sec. 4.4).

4.1 Protocol

The evaluation domain is the SULAND benchmark [40], which contains ground-level RGB images of PFM-1 and PMA-2 surface-laid landmines across the ITA and USA sites already introduced in Section 2. The real labeled baseline (**R1**) is a YOLOv11-L trained on the ITA split of SULAND, following the benchmark’s own recipe (details in Appendix A); it represents the conventional route in which labels collected at one site are reused for the next. All models, real and synthetic, are evaluated *once* on the USA split of SULAND (4,436 frames), which no model sees during training. This is a cross-site generalization setting: the terrain distribution, capture rig, and lighting all shift between train and test. Metrics are macro-F1 (**mF1**) over the two landmine classes, per-class recall, and mean average precision at intersection-over-union (IoU) 0.5 (**mAP50**). Every synthetic configuration is trained on 1,000 generated images with class balance 790:790 instances by design. We report mean $\pm$ standard deviation over five training seeds per configuration.

4.2 Generalization to real images

Table 1 lists the two baselines and the four levels of the mask-lock grade α . The reference synthetic configuration (**M2**, $\alpha=0.10$) reaches $\text{mF1} = 0.584 \pm 0.025$, against 0.393 ± 0.021 for the real labeled baseline (**R1**). The difference is several times the seed variability (≈ 5.8 combined seed standard deviations). We read this as preliminary evidence, with the caveats of Section 6: on this benchmark and under this recipe, the detector trained on our synthetic data generalized better across sites than the one trained on the larger real labeled set. R1’s absolute level also shows that the ITA-to-USA shift is severe, which contextualizes how much headroom the label-reuse route leaves on this benchmark. The raw frame counts overstate the supervision gap: of the 22,756 real training frames, 5,234 contain landmine instances, against roughly 900 of the 1,000 synthetic images, an effective ratio closer to 6:1 than to 23:1. One access asymmetry deserves note: R1 sees no data from the test site, whereas our pipeline consumes a handful of unlabeled test-site photographs (the exact budget is swept in Table 2), so the comparison is training on source-site labels alone versus unsupervised adaptation to the target site. Raw computer-graphics (CG) renders without the diffusion stage (**R2**) reach only 0.194 ± 0.111 : they teach the renderer, not the world, and their seed-to-seed variance is an order of magnitude larger than at any $\alpha > 0$.

4.3 Which curation choice matters

The mask-lock sweep in Table 1 shows an inverted-U across the four sampled values of α ; a denser grid would be needed to trace the curve between them.

Table 1: Cross-site generalization on SULAND (ITA / synthetic $\rightarrow$ USA split; five seeds). **R1** is the real labeled baseline, **R2** the raw computer-graphics (CG) pool, and **M0–M3** sweep the mask-lock grade α over synthetic pools of the same size. **Bold:** best value per column among the synthetic configurations; values within a tie band ($\Delta < 0.01$ on per-class recall, for which we do not report seed variance) are bolded together and should not be read as ordered.

	Config	mF1	R PFM-1	R PMA-2	mAP50
R1	Real labeled (22.7k)	0.393 ± 0.021	0.377	0.217	0.327 ± 0.015
R2	Raw CG, no diffusion	0.194 ± 0.111	0.208	0.185	0.136 ± 0.095
M0	Hard mask-lock ($\alpha=0$)	0.441 ± 0.051	0.579	0.196	0.346 ± 0.037
M1	Graded, $\alpha=0.05$	0.503 ± 0.060	0.571	0.285	0.401 ± 0.049
M2	Graded, $\alpha=0.10$ (<i>ref.</i>)	0.584 $\pm$ 0.025	0.577	0.418	0.481 $\pm$ 0.024
M3	Unlocked ($\alpha=1$)	0.466 ± 0.030	0.378	0.339	0.376 ± 0.034

Pixel-protecting the object (M0) preserves labels but also preserves the object’s appearance gap: PMA-2 recall stays at the real-data level (0.20) and mF1 does not clear the real baseline within error. Lightly diffusing the object (M2, $\alpha=0.10$) transfers the object’s appearance too: PMA-2 recall roughly doubles from 0.20 to 0.42, while PFM-1 recall stays flat at 0.57–0.58 across M0–M2. We speculate that the PFM-1’s distinctive silhouette already suffices for detection, whereas the low-contrast PMA-2 disc depends on surface appearance, which is exactly what the graded lock transfers. Removing the lock entirely (M3) dissolves parts of the object and gives back most of the gain. M3 also doubles as an unprotected-diffusion control, analogous in spirit to full-regeneration pipelines such as Károly and Galambos [15]: the LoRA-adapted SDEdit pass touches the whole frame, object included. In our runs M3 falls short of M2, suggesting that a substantial share of the gain comes from the graded lock rather than from the adapted diffusion pass alone. Among the axes we sweep, this one produces the largest shift.

Qualitative behavior of the lock. Figures 2 and 3 show scenes from the shared render pool across the mask-lock grade α , together with the scene-derived segmentation masks and 2D labels. Because the per-image diffusion seed is fixed, every column shows the *same* scene and the same diffusion draw; the only change is how much of the object the diffusion may touch. At $\alpha=0$ the object keeps its raw rendered appearance; at $\alpha=0.05$ – 0.10 the object inherits surface statistics from the terrain prior while its geometry and label are preserved; fully unlocked ($\alpha=1$), the diffusion dissolves the object into vegetation-like shapes and the label no longer marks a landmine. Figure 3 repeats the comparison on the MineInsight forest-floor terrain [23], using its RGB stream only, with the same methodology and the same grades, as a qualitative check that the behavior is not specific to grassland.

Table 2 sweeps three budget axes: reference photographs (1, 5, 10), patches per photograph (2, 5, 14), and generated image count (100, 250, 500, 1,000). The

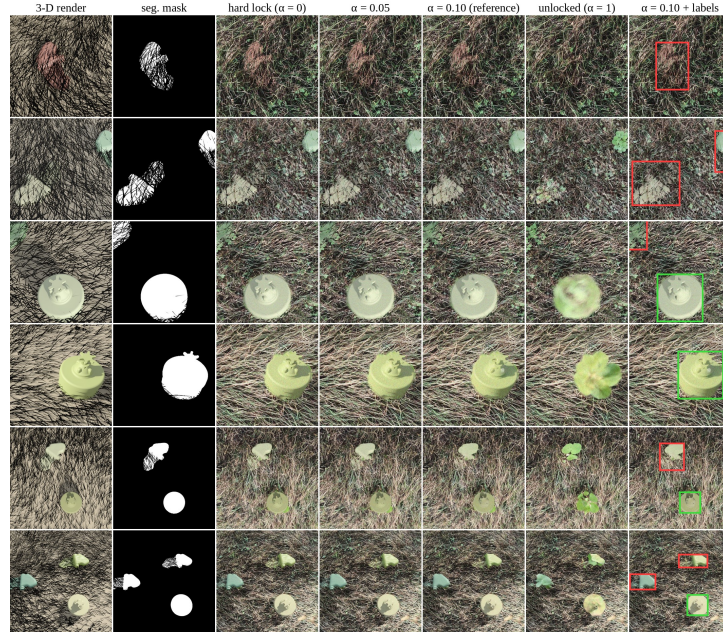


Fig. 2: Mask-lock grade α across six scenes (rows), same diffusion seed per row. Columns: 3D render, occlusion-aware segmentation mask, lock grades in increasing order ($\alpha = 0, 0.05, 0.10, 1$), and the reference output with its 2D labels (red: PFM-1, green: PMA-2). Rows 1–2: heavily occluded PFM-1s (41% and 51% visible); rows 3–4: PMA-2 close-ups; rows 5–6: multi-landmine scenes. Unlocked, objects are absorbed into the background prior, most completely where occlusion is heaviest.

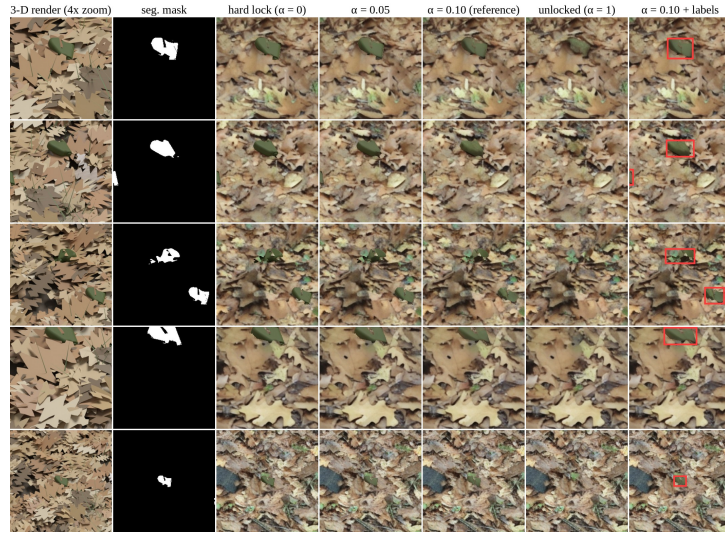


Fig. 3: The same comparison on the MineInsight forest-floor terrain (RGB stream; five scenes; leaf-litter LoRA), same columns as Figure 2; row 4 is a focal close-up at unchanged camera position. As on grassland, unconstrained diffusion absorbs the small object into the background prior; grades were evaluated quantitatively on SULAND only.

full sweep was run at the hard-lock setting ($\alpha=0$). The smallest-budget level of each axis (one photograph, two patches, one hundred images) was then re-measured at the reference lock ($\alpha=0.10$), five seeds each, as the most demanding test of the flatness claim; the remaining combinations were not re-run for compute reasons, as the second column of Table 2 shows. Within the measured cells, the photograph and patch axes are flat within error; notably, a *single* reference photograph at the reference lock reaches 0.562 ± 0.011 mF1, within one standard deviation of the ten-photograph configuration, and is the most seed-stable cell in the study. This presumes an approximately uniform terrain appearance across the site, as holds for the grassland evaluated here; we return to this assumption in Section 5. The image-count axis is the exception: at the reference lock the curve rises by ~ 0.07 mF1 from 100 to 1,000 images before flattening, still secondary to the $+0.14$ of the mask-lock axis itself. Within the ranges we could afford, quantity appears secondary to how the object is processed. A calibrated sensor-noise pass, ablated separately, leaves mF1 unchanged within seed variance at both lock settings (at $\alpha=0$: 0.441 vs. 0.448 ; at $\alpha=0.10$: 0.584 ± 0.025 vs. 0.571 ± 0.048), though we keep it for photometric fidelity of the generated images.

Table 2: Budget axes, five seeds per cell. Full sweep at hard lock ($\alpha=0$); axis extremes re-measured at the reference lock ($\alpha=0.10$); n/a = cell not re-measured. The reference configuration (10 photographs, 14 patches, 1,000 images) scores 0.441 ± 0.051 at $\alpha=0$ and 0.584 ± 0.025 at $\alpha=0.10$ (Table 1). **Bold:** best value per column among the rows shown.

	Axis, level	mF1 at $\alpha=0$	mF1 at $\alpha=0.10$
P1	1 photograph (14 crops)	0.485 ± 0.070	0.562 ± 0.011
P2	5 photographs	0.487 ± 0.028	n/a
C1	2 patches / photograph	0.461 ± 0.042	0.564 ± 0.023
C2	5 patches / photograph	0.505 ± 0.042	n/a
S1	100 generated images	0.396 ± 0.085	0.514 ± 0.042
S2	250 generated images	0.410 ± 0.084	n/a
S3	500 generated images	0.497 ± 0.051	n/a

4.4 Predictors of generalization

Every synthetic configuration in Tables 1 and 2 reached $\text{mAP}_{50} \geq 0.99$ on a validation split of the generated images that the detector never trains on (except the unlocked M3, which plateaued at 0.84–0.87, and one 100-image seed at 0.91), yet real-domain mF1 spanned 0.05 to 0.62 across configurations and seeds; as anticipated in Section 3, in-domain validation was uninformative about generalization in our runs. The spread is systematic rather than random: moving any configuration from the hard lock to the reference lock roughly halved its seed variance (*e.g.*, $0.070 \rightarrow 0.011$ for the one-photograph cell), so the graded lock appears to stabilize generalization as well as improve its mean. Within this study, curation decisions had to be guided by downstream evaluation.

5 Limitations

Five limitations bound the present findings. First, the scene originates from a single reference image, and the VLM’s first-pass reconstruction might be imprecise; an agent-in-the-loop refinement could reduce this imprecision. Second, the camera distance to the ground is only approximately recovered, while object scale matters for the graded lock. Third, the terrain is assumed approximately uniform, as in the grassland studied here and in the leaf-litter environment of Figure 3; real sites may mix vegetation with asphalt or bare regions, which would call for a semantic-region analysis and, eventually, a region-conditioned diffusion pass. Fourth, the study covers one target family, one evaluation split, and one detector, and deliberately does not ablate the 3D assets or their fidelity. Fifth, the mask-lock grade is sampled at four values only, so the inverted-U of Section 4.3 should be read as a trend rather than a fitted curve.

6 Conclusion and future work

We presented a preliminary study of a VLM-guided render-and-refine pipeline for detection in vegetated terrain when labeled real data are scarce, expensive, or dangerous to collect. The pipeline turns a handful of unlabeled photographs of the site and a small library of 3D meshes into a labeled detection dataset with free bounding boxes, masks, and per-instance occlusion. A graded mask-lock parameter interpolates between hard label protection and unconstrained diffusion of the object; in our experiments its middle regime generalized best. At that setting, a detector trained on our synthetic data matched or exceeded the same detector trained on a larger labeled dataset of real images for cross-site generalization on the SULAND benchmark [40], with comparable seed variance. If these findings hold beyond this benchmark, the per-site data question for detection pipelines of this kind would shift from collecting and labeling new frames to photographing the site and regenerating the training set.

Scope of the claim. These are preliminary findings on one target class family, one terrain type, one evaluation split, and one detector architecture. We also deliberately do not ablate the use of 3D meshes or the fidelity of the meshes themselves in the render. The synthetic-versus-real comparison should therefore be read as evidence that a small, curated synthetic set can be competitive for cross-site generalization under this recipe, not as a general claim about synthetic data superiority.

Future work. Four directions follow from the limitations we observed. *Richer 3D curation:* the VLM currently emits only minimal scene primitives, and substantial work remains on capturing the initial 3D distribution of a site inside the simulator; supporting the capture with range sensing (*e.g.*, a LiDAR back-support) could standardize the scene geometry and remove the dependence on any particular VLM, which we cannot meaningfully ablate here. *Better blending:* the mask-lock stage today paints the rendered object back through a binary visible-surface mask; soft harmonization and learned compositing could remove the small residual seam that remains at low α and may absorb the appearance gap the graded lock currently transfers. *Higher-fidelity meshes:* the object realism is upper-bounded by the mesh textures, especially for weathering, dirt, and paint wear that are absent from our meshes; capturing physically-based materials from inert ordnance would move the object closer to the target distribution without further diffusion pressure. *Broader evaluation:* a second detector family, additional terrain pairs (forest, sand, mixed), a denser mask-lock grid, and a cheap and calibrated predictor of generalization are natural next steps.

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the initial human-written manuscript for improved readability, clarity, and flow; all scientific content and conclusions remain the responsibility of the authors.

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A Appendix

A.1 Configuration

Generation. Blender (Cycles) renders at 960×960 px, 48 samples; 1–3 landmines per scene with a stratified 1:1 class draw (4 PFM-1 mesh variants, 1 PMA-2), 10% landmine-free negative frames; camera poses are rejected unless every frustum corner lands on vegetated ground. The draw balances classes over the whole pool: 900 of the 1,000 frames carry 1,580 landmine instances in total, 790 per class, and the remaining 100 frames are negatives. Per-instance visibility is computed from paired visible/full silhouette renders (pool median 0.96; 11% of instances below 0.70). The terrain LoRA uses rank 12, 1,000 steps, AdamW at 10^{-4} , 512-px crops, and the fixed caption “*sulandterrain field ground terrain, green and dry grass, clover, top-down photograph*”, reused verbatim at generation time. SDEdit runs at strength 0.42, 35 steps, classifier-free guidance 6.0, LoRA scale 0.8; the per-image noise generator is seeded with 42 + image id, which makes every ablation cell pixel-aligned to every other.

Table 3: Complete detector-training configuration (SULAND recipe, identical for all runs) and generation constants.

Training		Augmentation (SULAND)	
model	YOLOv11-L	mosaic	1.0
epochs	100	hsv (h/s/v)	0.015 / 0.7 / 0.4
patience	25	translate	0.1
input size	640	scale	0.5
batch	32	horizontal flip	0.5
optimizer	AdamW	mixup / copy-paste	0 / 0
initial lr	3×10^{-4} (cosine)	degrees / shear	0 / 0
weight decay	5×10^{-4}	deterministic mode	on
dropout	0.1	seeds	{42, 1, 2, 3, 4}
Generation			
SDEdit strength	0.42	LoRA rank / steps	12 / 1,000
diffusion steps	35	LoRA lr / crops	10^{-4} / 512 px
guidance	6.0	noise inject σ	15.8 (measured)
LoRA scale	0.8	negatives	10%

VLM scene generation. The scene-generation backend is Claude Fable 5 (Anthropic; model id `claude-fable-5`). A single prompt provides the reference photograph and asks the model to emit Python code that instantiates the terrain in Blender as scene primitives (ground, vegetation cards, clutter); no pre-owned asset library is involved. The initial fixed prompt, also shown in Figure 1, is: “Write a Blender Python script to build this terrain’s geometry using primitives, and match the camera orientation. Ignore any field objects.”

Capture realism. Per-scene sun strength follows the reference photograph’s mean luminance ($e_0 \cdot 2.5 (0.6 + 1.4\bar{L})$), and each render is gamma-matched to its own reference photograph’s mean luminance (median $\gamma = 0.42$). The sensor-noise target measured on the reference photographs is $\sigma = 12.3$ (0–255 scale); closed-loop calibration of the injected noise against the final JPEG (quality 88) converged at an injection of $\sigma = 15.8$, leaving a final synthetic statistic of 12.6.

Detector training. Table 3 lists the complete training configuration: the SULAND benchmark recipe, used unchanged for the real baseline and for every synthetic configuration. All rendering, LoRA fine-tuning, diffusion, and detector training ran on a single NVIDIA RTX 5090 (32 GB).

A.2 Quality-metric probes

We also probed two supporting signals, not used to guide the study but included here as sanity checks: (i) image-level distributional realism, measured as Kernel Inception Distance (KID) [3] between each generated set and real target-domain crops, gave a Spearman correlation of $\rho = -0.19$ ($p = 0.60$) against downstream

mF1 across configurations; it isolates the raw-CG pool as an outlier but does not rank the diffused sets. (ii) An open-vocabulary detection probe (Grounding-DINO [20] recall on the target class) reported near-saturated recall (0.88 to 0.98) for every generated set, including the raw-CG pool that generalizes worst ($\rho = -0.39$, $p = 0.27$). Both are observations of this study’s setting; we did not tune either probe, and different prompts, thresholds, or backbones could behave differently.

A.3 SULAND protocol details

The ITA split provides 22,756 labeled training frames; the USA split used for evaluation provides 4,436 frames. Evaluation uses the standard Ultralytics validation routine at IoU 0.5, run once per trained model; per-class F1 is read at the F1-optimal operating point of the precision–recall curve, computed identically for all models, and macro-F1 is the mean of the two per-class F1 scores.

A.4 Real-baseline provenance

The real baseline is trained on the SULAND ITA split: 22,756 frames, of which 5,234 contain landmine instances (5,510 boxes, $\approx 1:1$ between classes), by the same training configuration as above. The ten unlabeled adaptation photographs are drawn from the USA-site (evaluation) distribution; each was matched to its source evaluation frame (feature cosine > 0.95 , one per capture sequence). Removing the matched frames from the evaluation set is a null effect on the model the concern applies to, the synthetic configuration whose LoRA saw the photographs: the reference M2 scores mF1 0.584 ± 0.025 and mAP50 0.481 ± 0.024 on the full set, versus 0.584 ± 0.025 and 0.482 ± 0.024 with the matched frames removed (five seeds). Evaluation therefore uses the full set, and the overlap is disclosed rather than excluded. Frames from the same capture sequences share appearance with the adaptation photographs by construction; that is the site-adaptation setting itself rather than duplicate leakage, which the exclusion above bounds.